

Flipkart_XGBoost_Sentiment_Analysis

1. Abstract

Sentiment Analysis is a Natural Language Processing (NLP) technique used to determine whether a piece of text expresses a positive, negative, or neutral opinion. With the rapid growth of e-commerce platforms, customer reviews have become an important source of information for understanding customer satisfaction and product quality.

This project develops a sentiment analysis system using the Flipkart Product Customer Reviews Dataset. Text reviews are preprocessed and transformed into numerical features using TF-IDF vectorization. An XGBoost classifier is trained to predict customer sentiment from review text. The project demonstrates the complete machine learning pipeline from data preprocessing to model evaluation.

2. Introduction

Online shopping platforms such as Flipkart receive thousands of customer reviews every day. These reviews provide valuable insights into customer experiences and product quality.

However, manually reading and analyzing large numbers of reviews is difficult and time-consuming. Sentiment Analysis helps automate this process by identifying whether customer opinions are positive or negative.

Machine Learning and NLP techniques make it possible to build systems that can understand textual reviews and classify sentiments automatically.

3. Problem Statement

E-commerce companies need efficient ways to understand customer feedback from large volumes of product reviews.

The objective of this project is to build a machine learning model capable of automatically classifying customer reviews into sentiment categories based on review text.

4. Objectives

The main objectives of this project are:

- To understand sentiment analysis techniques.
 - To preprocess customer review text.
 - To convert textual data into numerical features.
 - To train an XGBoost classification model.
 - To evaluate model performance.
 - To predict customer sentiment automatically.
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5. Dataset Description

Dataset Name:

Flipkart Product Customer Reviews Dataset

Source:

<https://www.kaggle.com/datasets/niraliivaghani/flipkart-product-customer-reviews-dataset>

Dataset Description:

The dataset contains customer reviews and ratings for products available on Flipkart.

Typical Features:

Feature	Description
Product Name	Name of product
Review	Customer review text
Rating	Customer rating
Sentiment	Positive / Negative Label

Dataset Characteristics:

- Real-world customer reviews
 - Textual review data
 - Suitable for NLP applications
 - Useful for sentiment classification
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6. Machine Learning Workflow

The project follows a standard machine learning workflow:

1. Data Collection
 2. Data Exploration
 3. Data Cleaning
 4. Text Preprocessing
 5. Feature Extraction
 6. Dataset Splitting
 7. Model Training
 8. Model Evaluation
 9. Sentiment Prediction
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7. Exploratory Data Analysis (EDA)

EDA is performed to understand the dataset before training.

Analysis includes:

- Number of reviews
- Distribution of sentiments
- Rating distribution
- Most common review words
- Review length analysis

Benefits:

- Understand dataset characteristics
 - Detect class imbalance
 - Identify missing values
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8. Data Preprocessing

Raw review text cannot be directly used for machine learning.

The following preprocessing steps are applied:

1. Lowercase Conversion

All text is converted to lowercase.

Example:

Before:

"This Product Is Amazing"

After:

"this product is amazing"

2. Removal of Punctuation

Special characters are removed.

Examples:

! @ # \$ % ^ & * ()

3. Tokenization

Reviews are split into individual words.

Example:

"This phone is excellent"

becomes

["this", "phone", "is", "excellent"]

4. Stop Word Removal

Common words with little meaning are removed.

Examples:

- the
- is
- am
- are
- and
- of

5. Stemming

Words are reduced to their root form.

Examples:

liked → like

playing → play

working → work

9. Sentiment Label Creation

Customer ratings are converted into sentiment labels.

Example:

Rating	Sentiment
4–5	Positive
1–2	Negative
3	Neutral (optional)

This converts the problem into a classification task.

10. Feature Extraction

Machine Learning algorithms require numerical input.

Text reviews are transformed into numerical vectors using:

TF-IDF Vectorization

TF-IDF stands for:

Term Frequency – Inverse Document Frequency

Purpose:

- Measures word importance.
- Reduces influence of common words.
- Improves model performance.

Example:

Review:

"Excellent battery life"

becomes a numerical feature vector.

11. Dataset Splitting

The dataset is divided into:

Dataset	Percentage
Training Data	80%
Testing Data	20%

Training Data:

Used to train the model.

Testing Data:

Used to evaluate model performance.

12. Machine Learning Algorithm

XGBoost Classifier

XGBoost (Extreme Gradient Boosting) is a powerful ensemble learning algorithm widely used in machine learning competitions and real-world applications.

Reasons for Using XGBoost:

- High accuracy
- Fast training
- Handles large datasets efficiently
- Good performance on classification tasks
- Robust against overfitting

Working Principle:

XGBoost builds multiple decision trees sequentially. Each new tree corrects errors made by previous trees, resulting in improved prediction performance.

13. Model Training

The XGBoost model is trained using:

- TF-IDF feature vectors
- Customer review text
- Training dataset

The model learns patterns associated with positive and negative sentiments.

Examples:

Positive Keywords:

- excellent
- amazing
- good
- perfect
- satisfied

Negative Keywords:

- poor
 - bad
 - worst
 - damaged
 - disappointed
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14. Model Evaluation

The trained model is evaluated using testing data.

Evaluation Metrics

Accuracy

Measures overall correctness.

Accuracy = Correct Predictions / Total Predictions

Precision

Measures how many predicted positive sentiments are actually positive.

Recall

Measures how many actual positive sentiments are successfully detected.

F1-Score

Balances Precision and Recall.

Higher values indicate better performance.

15. Confusion Matrix

A confusion matrix summarizes prediction results.

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

Where:

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

Benefits:

- Identifies classification errors
- Helps evaluate model effectiveness

16. Results

The XGBoost model successfully classified customer reviews into sentiment categories.

Typical Results:

- Accuracy: 85% – 95%
- High Precision
- High Recall
- Strong F1-Score

The model effectively learned patterns from customer reviews and accurately predicted sentiment.

17. Advantages

- Automatic sentiment detection
 - Fast review analysis
 - Supports business decision-making
 - Scalable to large datasets
 - Improves customer feedback analysis
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18. Limitations

- Cannot fully understand sarcasm.
 - Performance depends on review quality.
 - May struggle with mixed sentiments.
 - Requires retraining on new data.
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19. Applications

Sentiment analysis can be used in:

- E-commerce review analysis
 - Product recommendation systems
 - Customer feedback monitoring
 - Brand reputation analysis
 - Social media analytics
 - Market research
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20. Future Enhancements

Possible improvements include:

- Multi-class sentiment classification
- Deep Learning Models
- LSTM Networks
- BERT-Based Sentiment Analysis
- Real-Time Sentiment Monitoring
- Deployment as a Web Application

21. Conclusion

This project successfully demonstrates the application of Machine Learning and Natural Language Processing for customer sentiment analysis. The Flipkart Product Customer Reviews Dataset was preprocessed and transformed into numerical features using TF-IDF vectorization. An XGBoost classifier was trained to classify customer reviews into sentiment categories. The model achieved strong performance and provided valuable insights into customer opinions. This project serves as an excellent beginner-level introduction to sentiment analysis, text mining, and machine learning-based classification systems.